



# OPTIMIZING DELIVERY ETAS WITH GRAPH-BASED NETWORK INTELLIGENCE

## BACKGROUND

Delhivery is India's largest fully-integrated logistics provider, operating a vast network of facilities, inter-city routes, and last-mile delivery across every major state. At the core of its operations is a hub-and-spoke model: shipments travel from a source facility through one or more intermediate hubs before reaching the destination, each hop is a segment of a multi-leg journey.

To estimate delivery times, Delhivery uses OSRM, a standard routing engine that assumes clean traffic and shortest paths. But real-world logistics is far messier: congestion, facility dwell time, seasonal volume spikes, and route-type constraints all cause actual delivery times to deviate significantly from predictions.

## THE CHALLENGE

The strategic question: Delhivery's OSRM system underestimates actual delivery time on a significant fraction of routes. Can a graph-based model, one that treats the logistics network as a connected graph of facilities and corridors, not a collection of independent point-to-point estimates, produce more accurate ETAs and identify which corridors and hubs are systematically causing delays?

When ETA is wrong, SLAs are missed and customers are unhappy

Downstream capacity planning breaks down across the network

There is currently no systematic way to identify which hubs and corridors are the biggest contributors to delays

Route-type decisions (FTL vs Carting) are made without accounting for graph position or structural risk of a facility

## YOUR MISSION

As a data science team, your goal is to build a graph-based intelligence system for Delhivery's logistics network. You will model the entire network as a directed graph, facilities as nodes, corridors as edges, and use this structure to produce smarter ETA predictions, surface bottleneck hubs, and generate actionable recommendations for the Head of Network Operations.

Your final output is not just a model, it is a consulting deliverable: a strategy memo that a real operations leader could act on.

## WHAT WE EXPECT FROM YOU

- Technical rigor, Clean, reproducible code with clear documentation. Model choices should be justified, not just applied
- Analytical thinking, Go beyond running the model. Interpret what the graph metrics mean for real operations. Why is a particular hub a bottleneck? What would fixing it cost vs. save?
- Business communication, The strategy memo must be written for an operations leader, not a data scientist. No raw model outputs, translate findings into decisions
- Creativity, Suggest additional insights, visualizations, or analyses beyond the provided scope if you spot something interesting in the data



## KEY TASKS AND DELIVERABLES

1. Graph construction & data pipeline - Parse and merge raw trip segments into a directed weighted graph where edge weights capture the median actual-vs-OSRM delay ratio per corridor, stratified by route type and time of day. Code must be clean, reproducible, and model choices justified - not just applied.
2. Bottleneck & corridor audit - Compute betweenness centrality, in/out-degree, and clustering coefficients to identify critical chokepoint hubs and chronically delayed corridors (actual time exceeds OSRM by >20%). Visualize and rank by SLA breach contribution.
3. Graph-enhanced ETA prediction model - Build and benchmark a baseline regression (trip-level features) against a GraphSAGE/node2vec-enhanced model. The graph model must demonstrably outperform the baseline on MAE and on % of trips with predicted ETA within 15% of actual. The "graph advantage" must be measured, not claimed.
4. FTL vs Carting decision framework - Build an ML-backed framework for route-type selection with the time-cost trade-off quantified for different corridor profiles, accounting for distance, time of day, and the source facility's graph position.
5. Network Operations Strategy Memo - Written for an operations leader, not a data scientist. Name the top 5 bottleneck hubs with their estimated SLA breach contribution, recommend corridor-specific interventions (parallel route, facility upgrade, or route-type shift), and quantify the % reduction in late deliveries and revenue-at-risk recovered if the top 3 hubs are upgraded. No raw model outputs - translate findings into decisions.

## EXPECTED IMPACT

- By the end of this project, your solution should help Delhivery's operations team:
- Make smarter ETA predictions that reduce the gap between promised and actual delivery time
- Identify and prioritize hub upgrades based on their structural risk and SLA breach contribution
- Reduce SLA breaches on chronic delay corridors through targeted interventions
- Improve route-type decisions with a data-backed FTL vs Carting framework
- Recover revenue at risk by quantifying and acting on the highest-impact bottlenecks

## WHAT YOUR FINAL SUBMISSION SHOULD INCLUDE

1. Well-documented code, data pipeline, graph construction, model training, and benchmarking
2. Graph visualizations of the logistics network with bottleneck hubs and delay corridors highlighted
3. Model performance comparison (baseline vs. graph-enhanced) on both MAE and the 15%-accuracy business metric
4. FTL vs Carting decision framework with supporting analysis
5. Network Operations Strategy Memo (1-2 pages) with the top 5 bottleneck hubs, corridor-specific interventions, and revenue impact estimates
6. Optional: A live Streamlit dashboard showing the network with real-time delay risk scores

## POINT OF CONTACT

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## DATASET & RESOURCES

[LINK \(DATASET\)](#)

[LINK \(RESOURCES\)](#)